3. In a three-digit number, when the tens and the hundreds digit are interchanged the new number is 54 more than three times the original number. If 198 is added to the number, the digits are reversed. The tens digit exceeds the hundreds digit by twice as that of the tens digit exceeds the unit digit. Find the original number.
4. Find the least common multiple of $xy(k^2+1)+k(x^2+y^2)$ and $xy(k^2-1)+k(x^2-y^2)$
5. Find the GCD of the following by division algorithm

   $2x^4+13x^3+27x^2+23x+7$, x^3+3x^2+3x+1, x^2+2x+1
6. Reduce the given Rational expressions to its lowest form

   (i) $\dfrac{x^{3a}-8}{x^{2a}+2x^a+4}$ (ii) $\dfrac{10x^3-25x^2+4x-10}{-4-10x^2}$
7. Simplify $\dfrac{\dfrac{1}{p}+\dfrac{1}{q+r}}{\dfrac{1}{p}-\dfrac{1}{q+r}}\times\left(1+\dfrac{q^2+r^2-p^2}{2qr}\right)$
8. Arul, Madan and Ram working together can clean a store in 6 hours. Working alone, Madan takes twice as long to clean the store as Arul does. Ram needs three times as long as Arul does. How long would it take each if they are working alone?
9. Find the square root of $289x^4-612x^3+970x^2-684x+361$.
10. Solve $\sqrt{y+1}+\sqrt{2y-5}=3$
11. A boat takes 1.6 hours longer to go 36 kms up a river than down the river. If the speed of the water current is 4 km per hr, what is the speed of the boat in still water?
12. Is it possible to design a rectangular park of perimeter 320 m and area $4800\ m^2$? If so find its length and breadth.
13. At t minutes past 2 pm, the time needed to 3 pm is 3 minutes less than $\dfrac{t^2}{4}$. Find t.
14. The number of seats in a row is equal to the total number of rows in a hall. The total number of seats in the hall will increase by 375 if the number of rows is doubled and the number of seats in each row is reduced by 5. Find the number of rows in the hall at the beginning.
15. If α and β are the roots of the polynomial $f(x)=x^2-2x+3$, find the polynomial whose roots are (i) $\alpha+2$, $\beta+2$ (ii) $\dfrac{\alpha-1}{\alpha+1}$, $\dfrac{\beta-1}{\beta+1}$.
16. If –4 is a root of the equation $x^2+px-4=0$ and if the equation $x^2+px+q=0$ has equal roots, find the values of p and q.
17. Two farmers Thilagan and Kausigan cultivates three varieties of grains namely rice, wheat and ragi. If the sale (in ₹) of three varieties of grains by both the farmers in the month of April is given by the matrix.

10th_Maths_Chapter 3_English.indd 157 12/23/2024 5:35:22 PM